

Anisakis simplex: from Obscure Infectious Worm to Inducer of Immune Hypersensitivity

Summary: Infection of humans with the nematode worm parasite *Anisakis simplex* was first described in the 1960s in association with the consumption of raw or undercooked fish. During the 1990s it was realized that even the ingestion of dead worms in food fish can cause severe hypersensitivity reactions, that these may be more prevalent than infection itself, and that this outcome could be associated with food preparations previously considered safe. Not only may allergic symptoms arise from infection by the parasites ("gastroallergic anisakiasis"), but true anaphylactic reactions can also occur following exposure to allergens from dead worms by food-borne, airborne, or skin contact routes. This review discusses *A. simplex* pathogenesis in humans, covering immune hypersensitivity reactions both in the context of a living infection and in terms of exposure to its allergens by other routes. Over the last 20 years, several studies have concentrated on *A. simplex* antigen characterization and innate as well as adaptive immune response to this parasite. Molecular characterization of *Anisakis* allergens and isolation of their encoding cDNAs is now an active field of research that should provide improved diagnostic tools in addition to tools with which to enhance our understanding of pathogenesis and controversial aspects of *A. simplex* allergy. We also discuss the potential relevance of parasite products such as allergens, proteinases, and proteinase inhibitors and the activation of basophils, eosinophils, and mast cells in the induction of *A. simplex*-related immune hypersensitivity states induced by exposure to the parasite, dead or alive.